SkillNavora

Supplement B

Missing Excel, SQL, business, governance and production topics

Fills the gaps in course Modules 1, 2, 6, 7 and 13, and the governance and business modules of the prompt.

Course gap check: supplement to the Data Science, AI and Data Engineering course. Existing lessons are unchanged.

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Supplement B: Gap topics for the existing modules

Topics the audit marks as partial or missing, written for the same learner and in the same structure as Supplement A.

Contents

- B1 · Excel Automation and Missing Excel Skills (course Module 1 gaps)
- B2 · SQL Extras for Analysts and Data Engineers (course Module 2 gaps)
- B3 · Business Problem-Solving and Analytics Consulting (course Module 16 gap)
- B4 · Requirements, Communication and Delivery
- B5 · Data Governance, Privacy and Responsible AI (course Module 15 gap)
- B6 · Production Python for Automation (course Module 6 gaps)
- B7 · Recommendation Systems (course Module 7 gap)
- B8 · APIs, OpenAPI, OAuth and Integration Depth (Module 13 gaps)
- B9 · Linux and the Command Line (Part C gap)
- B10 · Technical Documentation and Production Troubleshooting (Part C gap)

B1 Excel Automation and Missing Excel Skills (course Module 1 gaps)

B1.1

Flash Fill, Text to Columns and standardising data

WHAT IS IT?

> Flash Fill detects a pattern from examples and fills the rest; Text to Columns splits one column into several using a delimiter or fixed width.

WHY IS IT IMPORTANT?

> Analysts spend much of their time cleaning data; these tools remove hours of manual work.

WHERE IS IT USED IN INDUSTRY?

Used in finance, HR and operations reporting to clean exports from systems.

HOW DOES IT WORK?

- Type the result for the first row, then press Ctrl+E (Flash Fill).
- Data > Text to Columns splits by comma, space or fixed width.
- Standardise case with PROPER, UPPER, LOWER and TRIM.

PRACTICAL EXAMPLE

> Full names 'Asha Rao' split into first and last name; phone numbers standardised to +91 format.

TOOL OR SOFTWARE

Microsoft Excel (Windows or Microsoft 365 web)

HANDS-ON EXERCISE

Clean a list of 200 customer names and phone numbers; document each step.

COMMON MISTAKES

Using Flash Fill on data with mixed patterns without checking the result.

INTERVIEW QUESTION

How would you split a column of 'City, State' values in Excel?

OFFICIAL AND FURTHER READING

- Microsoft: Flash Fill (official)

B1.2

Excel macros and VBA: recording and writing automation

WHAT IS IT?

> A macro is a recorded or written set of steps. VBA (Visual Basic for Applications) is the language behind macros in Excel.

WHY IS IT IMPORTANT?

> Repetitive monthly reports can run in seconds instead of hours, and a macro can be checked and reused.

WHERE IS IT USED IN INDUSTRY?

Common in finance, operations and back-office teams.

HOW DOES IT WORK?

- Developer tab > Record Macro, perform the steps, stop recording.
- Alt+F11 opens the VBA editor to read or write code.
- Save macro-enabled files as .xlsm and sign or review macros from unknown sources.

PRACTICAL EXAMPLE

> A macro formats a monthly sales sheet, removes blank rows and saves it as a PDF.

TOOL OR SOFTWARE

Microsoft Excel VBA editor

HANDS-ON EXERCISE

Record a macro that formats a table and then edit its code to add a loop over three sheets.

COMMON MISTAKES

Recording macros without reading the code; hard-coded cell addresses break when data moves.

INTERVIEW QUESTION

What is the risk of running macros from an unknown file?

OFFICIAL AND FURTHER READING

- Microsoft VBA documentation (official)

B1.3

Office Scripts and AI-assisted Excel formulas

WHAT IS IT?

> Office Scripts are TypeScript automations for Excel on the web. AI tools can draft formulas, but each formula must be checked.

WHY IS IT IMPORTANT?

> Office Scripts run in Microsoft 365 without desktop macros, and can be triggered by Power Automate.

WHERE IS IT USED IN INDUSTRY?

Used in Microsoft 365 teams to automate reports that refresh on a schedule.

HOW DOES IT WORK?

- Automate > New Script in Excel on the web.
- Scripts can be run by Power Automate on a schedule.
- AI-written formulas are tested on a sample before use.

PRACTICAL EXAMPLE

> A script creates a summary sheet each Monday; Power Automate runs it and emails the file.

TOOL OR SOFTWARE

Excel on the web, Power Automate

HANDS-ON EXERCISE

Write an Office Script that adds a totals row and run it.

COMMON MISTAKES

Trusting an AI formula without checking edge cases such as blanks and errors.

INTERVIEW QUESTION

What is an Office Script and how does it differ from VBA?

OFFICIAL AND FURTHER READING

- Office Scripts documentation (official)

B1.4

Excel business projects from the prompt (not yet in the course)

WHAT IS IT?

> Projects: customer complaint analysis, financial performance tracker, employee productivity dashboard, customer segmentation and sales performance dashboard.

WHY IS IT IMPORTANT?

> Portfolio projects show the skills to employers, with real business questions.

WHERE IS IT USED IN INDUSTRY?

Analyst roles in every industry.

HOW DOES IT WORK?

- Brief: business question, data, deliverable, metric.
- Clean, model, analyse, visualise, recommend.
- Write a one-page summary for a manager.

PRACTICAL EXAMPLE

> Complaint analysis: count complaints by category and month, flag repeat customers, recommend two fixes with expected impact.

TOOL OR SOFTWARE

Excel

HANDS-ON EXERCISE

Complete the complaint analysis brief with a sample dataset you create.

COMMON MISTAKES

Dashboards with no question or recommendation.

INTERVIEW QUESTION

How would you analyse customer complaints for a telecom company?

OFFICIAL AND FURTHER READING

- Microsoft Excel training (official)

B2 SQL Extras for Analysts and Data Engineers (course Module 2 gaps)

B2.1

Recursive CTEs and hierarchies

WHAT IS IT?

> A recursive CTE repeats a query until no new rows appear, used for hierarchies such as organisation charts or bill-of-materials.

WHY IS IT IMPORTANT?

> Hierarchical data appears in HR, product and supply chain systems.

WHERE IS IT USED IN INDUSTRY?

Org charts, category trees and supply chains.

HOW DOES IT WORK?

- WITH RECURSIVE tree AS (anchor UNION ALL recursive step).
- The recursive step joins the CTE to the table again.
- Always include a stop condition to avoid endless loops.

PRACTICAL EXAMPLE

> WITH RECURSIVE reports AS (SELECT id, manager_id, 1 AS level FROM staff WHERE manager_id IS NULL UNION ALL SELECT s.id, s.manager_id, r.level+1 FROM staff s JOIN reports r ON s.manager_id = r.id) SELECT * FROM reports;

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Build the reporting-level query for a staff table of 20 people.

COMMON MISTAKES

Forgetting the stop condition.

INTERVIEW QUESTION

How would you find everyone who reports to a manager at any depth?

OFFICIAL AND FURTHER READING

- PostgreSQL WITH queries (official)

B2.2

Percentiles, medians and distribution queries

WHAT IS IT?

> Percentiles show the value below which a given share of rows fall. PERCENTILE_CONT interpolates; PERCENTILE_DISC picks an actual value.

WHY IS IT IMPORTANT?

> Medians and percentiles describe skewed business data (order values, delivery times) better than averages.

WHERE IS IT USED IN INDUSTRY?

Used for SLA reporting and pricing analysis.

HOW DOES IT WORK?

- PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY amount) is the median in PostgreSQL.
- Check the database's exact syntax; platforms differ.

PRACTICAL EXAMPLE

> The 90th percentile of delivery time per city, to report SLA breaches.

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Compute median and 90th percentile order value per region on a sample of 10,000 orders.

COMMON MISTAKES

Using AVG for skewed data without checking the median.

INTERVIEW QUESTION

Why report the median rather than the mean for delivery times?

OFFICIAL AND FURTHER READING

- PostgreSQL aggregate functions (official)

B2.3

Business SQL: cohort, funnel, churn, CLV and retention

WHAT IS IT?

> Cohort analysis groups customers by first purchase month and tracks their activity over time. Funnel analysis counts users at each step. Churn is the share lost in a period. CLV (customer lifetime value) estimates profit per customer over time.

WHY IS IT IMPORTANT?

> These are the core metrics in e-commerce, SaaS, banking and telecom analytics.

WHERE IS IT USED IN INDUSTRY?

Used by growth, marketing and customer success teams.

HOW DOES IT WORK?

- Cohort: first_month per customer, then count active customers by months since first purchase.
- Funnel: count distinct users per step, then divide.
- Churn: customers active last period and not this period, divided by active last period.
- CLV (simple): average order value x purchase frequency x expected years.

PRACTICAL EXAMPLE

> Cohort retention table: rows are first-purchase months, columns are months since first purchase, values are the share still buying.

TOOL OR SOFTWARE

PostgreSQL, sample e-commerce data

HANDS-ON EXERCISE

Build a cohort table for 12 months of sample orders and explain the drop after month 1.

COMMON MISTAKES

Counting orders instead of distinct customers.

INTERVIEW QUESTION

How do you calculate monthly churn with SQL?

OFFICIAL AND FURTHER READING

- Mode SQL tutorials: cohort analysis (official learning site)

B2.4

Incremental loading and data-quality SQL

WHAT IS IT?

> Incremental loading processes only new or changed rows using a watermark. Data-quality SQL tests check duplicates, nulls and orphans.

WHY IS IT IMPORTANT?

> Full reloads become too slow and expensive as data grows.

WHERE IS IT USED IN INDUSTRY?

Standard in data warehouse loading.

HOW DOES IT WORK?

- Store the last loaded timestamp.
- Select rows with updated_at greater than the watermark.
- Check: SELECT key, COUNT(*) FROM t GROUP BY key HAVING COUNT(*) > 1.

PRACTICAL EXAMPLE

> Load only orders updated since yesterday, then run a duplicate check on order_id.

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Build an incremental load with a watermark table and two quality checks.

COMMON MISTAKES

Using a watermark based on load time instead of source update time.

INTERVIEW QUESTION

How do you load only new data into a warehouse table?

OFFICIAL AND FURTHER READING

- PostgreSQL SQL reference (official)

B2.5

Database platforms: SQL Server, BigQuery, Snowflake and Databricks SQL

WHAT IS IT?

> The same SQL idea works on several platforms; each has its own functions, pricing and performance features.

WHY IS IT IMPORTANT?

> Job descriptions name platforms; you should know the differences in syntax, pricing model and tuning.

WHERE IS IT USED IN INDUSTRY?

Enterprises use SQL Server; cloud analytics teams use BigQuery, Snowflake or Databricks SQL.

HOW DOES IT WORK?

- SQL Server: T-SQL, SSMS, indexes and columnstore.
- BigQuery: pay per bytes scanned; partition and cluster tables.
- Snowflake: virtual warehouses, pay per compute time.
- Databricks SQL: SQL over Delta tables.

PRACTICAL EXAMPLE

> The same daily revenue query runs on BigQuery with partitioned tables to cut scan cost.

TOOL OR SOFTWARE

BigQuery sandbox (check current terms), Snowflake trial (check terms)

HANDS-ON EXERCISE

Run one query on two platforms and compare how they are priced and tuned.

COMMON MISTAKES

Writing SELECT * on pay-per-scan platforms.

INTERVIEW QUESTION

How does BigQuery pricing change how you write queries?

OFFICIAL AND FURTHER READING

- BigQuery documentation (official)
- Snowflake documentation (official)

B3 Business Problem-Solving and Analytics Consulting (course Module 16 gap)

B3.1

Business KPIs, models and process mapping

WHAT IS IT?

> A KPI is a measured result tied to a goal. A business model explains how money is made; process mapping shows each step of a workflow.

WHY IS IT IMPORTANT?

> Analysts who know the business choose the right metric and find the real problem.

WHERE IS IT USED IN INDUSTRY?

Consultants and analysts in banking, retail and healthcare start here.

HOW DOES IT WORK?

- Start from the goal, then the KPI, then the driver metrics.
- Map the process step by step with owners and times.
- Mark the bottleneck and the cost of delay.

PRACTICAL EXAMPLE

> Loan approval takes 9 days; mapping shows 5 days in document collection. The KPI is time to approval; the driver is documents received on day 1.

TOOL OR SOFTWARE

Miro or draw.io for process maps

HANDS-ON EXERCISE

Map one process you know, with three KPIs and one bottleneck.

COMMON MISTAKES

Choosing vanity metrics.

INTERVIEW QUESTION

What is the difference between a KPI and a driver metric?

OFFICIAL AND FURTHER READING

- Atlassian: process mapping (official guide)

B3.2

Problem discovery: stakeholder interviews, Five Whys and fishbone analysis

WHAT IS IT?

> Stakeholder interviews collect pain points. Five Whys asks why repeatedly to reach a root cause. A fishbone diagram groups causes into categories (people, process, tools, data).

WHY IS IT IMPORTANT?

> Solving the symptom wastes money; the root cause is what to fix.

WHERE IS IT USED IN INDUSTRY?

Used in operations, quality and consulting projects.

HOW DOES IT WORK?

- Interview 5 to 8 people across roles.
- Write a problem statement: who, what, impact, how often.
- Five Whys: ask why until the cause is a process or system you can change.
- Fishbone: list causes under each category.

PRACTICAL EXAMPLE

> Problem: late deliveries. Why 1: stock is out. Why 2: forecast is wrong. Why 3: sales data is a week late. Root cause: weekly data load.

TOOL OR SOFTWARE

Whiteboard, Miro

HANDS-ON EXERCISE

Run a Five Whys on a problem from your work and write the root cause.

COMMON MISTAKES

Stopping at the first answer.

INTERVIEW QUESTION

How would you find the root cause of rising returns?

OFFICIAL AND FURTHER READING

- ASQ: root cause analysis (official)

B3.3

Hypotheses, analytical questions and recommendations

WHAT IS IT?

A hypothesis is a testable statement. Analysis answers a question with data; a recommendation says what to do and what result to expect.

WHY IS IT IMPORTANT?

Recommendations that are not tied to a question and a measure are ignored.

WHERE IS IT USED IN INDUSTRY?

Consultants and analysts present recommendations to executives.

HOW DOES IT WORK?

- Write the question, then the hypothesis, then the data needed.
- Test it with the data; state what would prove it wrong.
- Recommend an action with an owner and a measure.

PRACTICAL EXAMPLE

Hypothesis: a reminder email reduces missed appointments by 15%. Test with an A/B split over four weeks.

TOOL OR SOFTWARE

Excel, SQL, Python

HANDS-ON EXERCISE

Write three hypotheses for a shop's falling sales and the data test for each.

COMMON MISTAKES

Recommendations without expected impact.

INTERVIEW QUESTION

How do you turn an analysis into a recommendation?

OFFICIAL AND FURTHER READING

- Harvard Business Review: data-driven decisions (learning article)

B4 Requirements, Communication and Delivery

B4.1

Requirements, user stories, acceptance criteria and PRDs

WHAT IS IT?

> Requirements say what the solution must do. User stories describe value from the user's view. Acceptance criteria say how to test it. A PRD (product requirements document) gathers these.

WHY IS IT IMPORTANT?

> Clear requirements stop rework and disputes with stakeholders.

WHERE IS IT USED IN INDUSTRY?

Product, analytics and BI teams write these every week.

HOW DOES IT WORK?

- Story: As a [role], I want [need], so that [benefit].
- Acceptance: Given [context], when [action], then [result].
- Non-functional requirements: speed, security, accuracy, availability.

PRACTICAL EXAMPLE

> As an operations manager, I want a daily late-order list, so that I can call customers before 10:00. Given an order is late by 2 hours, then it appears on the list by 08:00.

TOOL OR SOFTWARE

Jira, Confluence

HANDS-ON EXERCISE

Write five user stories with acceptance criteria for a dashboard.

COMMON MISTAKES

Stories without testable criteria.

INTERVIEW QUESTION

What is an acceptance criterion?

OFFICIAL AND FURTHER READING

- Atlassian: user stories (official guide)

B4.2

Communicating insights and data storytelling

WHAT IS IT?

> Data storytelling states the main message first, shows the evidence in one chart, and ends with the decision needed.

WHY IS IT IMPORTANT?

> Executives decide from short, clear messages. Long reports go unread.

WHERE IS IT USED IN INDUSTRY?

Analysts, FDEs and consultants present to leaders weekly.

HOW DOES IT WORK?

- Start with the answer.
- One chart per message.
- Name the decision, the owner and the date.

PRACTICAL EXAMPLE

> Slide: Churn rose 4 points in March, driven by two plans. Decision: fix onboarding for those plans by May.

TOOL OR SOFTWARE

PowerPoint, Power BI

HANDS-ON EXERCISE

Turn a 10-slide report into a 3-slide summary with a decision.

COMMON MISTAKES

Showing every chart on the dashboard.

INTERVIEW QUESTION

How would you present bad news to a manager?

OFFICIAL AND FURTHER READING

- Storytelling with Data (official site)

B4.3

Agile and project delivery for analytics teams

WHAT IS IT?

Agile delivers in short cycles with a backlog, sprints and reviews. Kanban shows flow on a board. Delivery management tracks scope, risk and dependencies.

WHY IS IT IMPORTANT?

Analytics work is often unclear at the start; short cycles show progress and adjust scope.

WHERE IS IT USED IN INDUSTRY?

Used by data and product teams across industries.

HOW DOES IT WORK?

- Backlog ordered by value.
- Sprint of 1 to 2 weeks with a goal.
- Review with stakeholders; retrospect and improve.

PRACTICAL EXAMPLE

A two-week sprint delivers the first version of a sales dashboard; the review finds two new metrics to add.

TOOL OR SOFTWARE

Jira, Trello

HANDS-ON EXERCISE

Plan a four-week analytics project in four sprints with goals.

COMMON MISTAKES

Planning without stakeholder reviews.

INTERVIEW QUESTION

How do you run an analytics project in sprints?

OFFICIAL AND FURTHER READING

- Scrum Guide (official)

B5 Data Governance, Privacy and Responsible AI (course Module 15 gap)

B5.1

Data ownership, stewardship and retention

WHAT IS IT?

> An owner decides who may use a dataset; a steward keeps it accurate. Retention sets how long data is kept.

WHY IS IT IMPORTANT?

> Clear ownership makes fixes happen; retention limits risk and cost.

WHERE IS IT USED IN INDUSTRY?

Required in banking, insurance and health data.

HOW DOES IT WORK?

- Each dataset has one owner and one steward.
- Retention period written in policy.
- Deletion runs on schedule.

PRACTICAL EXAMPLE

> Customer call recordings kept 90 days, then deleted by a scheduled job with a log.

TOOL OR SOFTWARE

Catalogue tools (see A4.3)

HANDS-ON EXERCISE

Write a data ownership sheet for five tables in a sales system.

COMMON MISTAKES

No owner for shared tables.

INTERVIEW QUESTION

Who should own customer data in a bank?

OFFICIAL AND FURTHER READING

- DAMA DMBOK overview (official body)

B5.2

Privacy: GDPR and India's Digital Personal Data Protection Act

WHAT IS IT?

> Privacy laws set rules for collecting, using, storing and deleting personal data. GDPR applies to EU residents' data; India's DPDP Act 2023 covers digital personal data in India.

WHY IS IT IMPORTANT?

> Breaches and fines follow poor privacy design. Analysts must minimise and protect personal data.

WHERE IS IT USED IN INDUSTRY?

Every customer-data project in India or Europe.

HOW DOES IT WORK?

- Collect only what is needed (data minimisation).
- Record consent and purpose.
- Give people access, correction and deletion rights where the law requires.
- Report breaches as the law requires.

PRACTICAL EXAMPLE

> A marketing dataset removes phone numbers not needed for the campaign and keeps a consent flag for each contact.

TOOL OR SOFTWARE

Data inventory sheet

HANDS-ON EXERCISE

Review a sample dataset and list the personal fields, purpose and retention for each.

COMMON MISTAKES

Copying production personal data into test files.

INTERVIEW QUESTION

What is data minimisation?

OFFICIAL AND FURTHER READING

- GDPR text (official EU site)
- MeitY: Digital Personal Data Protection Act (official)

B5.3

Data masking, row-level security and access control

WHAT IS IT?

> Masking hides sensitive values (for example showing the last four digits). Row-level security limits which rows a user sees.

WHY IS IT IMPORTANT?

> Analysts need data without exposing personal details.

WHERE IS IT USED IN INDUSTRY?

Used in banks, HR and healthcare analytics.

HOW DOES IT WORK?

- Mask in queries or in the database.
- Power BI and database RLS filter rows by user.
- Audit who accessed sensitive tables.

PRACTICAL EXAMPLE

> A regional manager sees only their region's rows in the Power BI report through RLS.

TOOL OR SOFTWARE

Power BI RLS, PostgreSQL row-level security

HANDS-ON EXERCISE

Set up a row filter for two regions and test it with two users.

COMMON MISTAKES

Relying on report-level hiding instead of security at the data layer.

INTERVIEW QUESTION

How do you stop an analyst from seeing salary data they should not see?

OFFICIAL AND FURTHER READING

- Power BI row-level security (official)

B5.4

Responsible AI, threat modelling and incident response

WHAT IS IT?

> Responsible AI covers fairness, transparency, human oversight and security. Threat modelling lists assets, threats and controls. Incident response plans how to act when something goes wrong.

WHY IS IT IMPORTANT?

> AI systems can leak data, be manipulated (prompt injection) or treat groups unfairly.

WHERE IS IT USED IN INDUSTRY?

Required for AI products in regulated sectors.

HOW DOES IT WORK?

- Check model performance by group.
- Threat model: data, prompts, tools, outputs.
- Plan detection, containment, fix and communication for incidents.

PRACTICAL EXAMPLE

> A chatbot leaks an internal price list through prompt injection; the incident plan disables the tool, rotates keys and updates guardrails.

TOOL OR SOFTWARE

OWASP Top 10 for LLM Applications

HANDS-ON EXERCISE

Write a threat model for an AI chatbot with five threats and controls.

COMMON MISTAKES

Treating guardrails as a complete solution.

INTERVIEW QUESTION

How would you test an AI assistant for prompt injection?

OFFICIAL AND FURTHER READING

- OWASP Top 10 for LLM Applications (official)
- NIST AI Risk Management Framework (official)

B6 Production Python for Automation (course Module 6 gaps)

B6.1

Automation: email, scheduled scripts and report generation

WHAT IS IT?

> Automation scripts run on a schedule, read files or APIs, produce reports and send them to people.

WHY IS IT IMPORTANT?

> Manual reports waste time and are error-prone.

WHERE IS IT USED IN INDUSTRY?

Common in every analytics team.

HOW DOES IT WORK?

- Use a scheduler (cron, Task Scheduler, Airflow).
- Send email with smtplib or a provider API; never store passwords in code.
- Log each run and alert on failure.

PRACTICAL EXAMPLE

> A script runs at 07:00, builds the daily sales PDF and emails it to managers.

TOOL OR SOFTWARE

Python, smtplib, cron

HANDS-ON EXERCISE

Automate a daily CSV summary and log each run.

COMMON MISTAKES

Hard-coded passwords.

INTERVIEW QUESTION

How would you schedule a Python report and handle failures?

OFFICIAL AND FURTHER READING

- Python smtplib (official)
- Python logging (official)

B6.2

Logging, type hints, packaging and code organisation

WHAT IS IT?

> Logging records what a program does. Type hints document expected types and let tools catch mistakes. Packaging makes code installable.

WHY IS IT IMPORTANT?

> Production code must be readable, observable and reusable.

WHERE IS IT USED IN INDUSTRY?

Expected in data engineering and software roles.

HOW DOES IT WORK?

- logging.basicConfig with levels (INFO, WARNING, ERROR).
- def total(values: list[float]) → float.
- pyproject.toml defines the package.

PRACTICAL EXAMPLE

> A pipeline module logs row counts and fails with a clear message on bad input.

TOOL OR SOFTWARE

Python logging, mypy (check docs)

HANDS-ON EXERCISE

Add logging and type hints to one of your scripts and run mypy.

COMMON MISTAKES

Logging secrets.

INTERVIEW QUESTION

What should a production log include?

OFFICIAL AND FURTHER READING

- Python logging HOWTO (official)

B7 Recommendation Systems (course Module 7 gap)

B7.1

Recommendation systems: collaborative and content-based

WHAT IS IT?

> Recommenders suggest items. Collaborative filtering uses patterns across users; content-based filtering uses item features; hybrid systems combine both.

WHY IS IT IMPORTANT?

> Recommendations drive sales and engagement in e-commerce, streaming and banking.

WHERE IS IT USED IN INDUSTRY?

Retail, media and fintech product teams.

HOW DOES IT WORK?

- Build a user-item matrix.
- Measure similarity between users or items.
- Evaluate with precision at k and hit rate on held-out interactions.

PRACTICAL EXAMPLE

> Users who bought the same two products get a suggestion of a third product bought by most of them.

TOOL OR SOFTWARE

scikit-learn, pandas, Surprise library (open source)

HANDS-ON EXERCISE

Build an item-based recommender on a sample ratings file and report precision at 5.

COMMON MISTAKES

Recommending items the user already owns.

INTERVIEW QUESTION

How would you recommend products to a new user with no history?

OFFICIAL AND FURTHER READING

- scikit-learn user guide (official)
- Surprise library docs (official)

B7.2

Regression metrics reference: MAE, MSE, RMSE and R-squared

WHAT IS IT?

> MAE is the average absolute error; MSE the average squared error; RMSE its square root, in the same units as the target; R-squared the share of variation explained.

WHY IS IT IMPORTANT?

> Choosing the right error metric matches the business cost of mistakes.

WHERE IS IT USED IN INDUSTRY?

Forecasting and pricing teams.

HOW DOES IT WORK?

- MAE is robust to outliers; RMSE punishes large errors.
- R-squared can mislead when models are very flexible.

PRACTICAL EXAMPLE

> A delivery-time model has RMSE 2.5 days: typical error is about two and a half days.

TOOL OR SOFTWARE

scikit-learn metrics

HANDS-ON EXERCISE

Compute MAE, RMSE and R-squared for a regression and explain each to a manager.

COMMON MISTAKES

Using R-squared alone to judge a model.

INTERVIEW QUESTION

When would you prefer MAE over RMSE?

OFFICIAL AND FURTHER READING

- scikit-learn regression metrics (official)

B8 APIs, OpenAPI, OAuth and Integration Depth (Module 13 gaps)

B8.1

OpenAPI and Swagger: describing APIs

WHAT IS IT?

> OpenAPI is a standard file (YAML or JSON) that describes an API's endpoints, parameters and responses. Swagger UI shows it as interactive documentation.

WHY IS IT IMPORTANT?

> Integrators and AI tools read OpenAPI files to call APIs correctly.

WHERE IS IT USED IN INDUSTRY?

Every API-first company publishes OpenAPI specs.

HOW DOES IT WORK?

- Paths and methods.
- Parameters and request bodies.
- Responses and schemas.
- Security schemes.

PRACTICAL EXAMPLE

> An OpenAPI file describes GET /orders?status=open and its response schema.

TOOL OR SOFTWARE

Swagger Editor

HANDS-ON EXERCISE

Write an OpenAPI description for two endpoints of your own app.

COMMON MISTAKES

Specs that are not kept in sync with the code.

INTERVIEW QUESTION

What is OpenAPI used for?

OFFICIAL AND FURTHER READING

- OpenAPI specification (official)

B8.2

OAuth 2.0 in depth and secure tokens

WHAT IS IT?

> OAuth 2.0 lets an app act for a user without the user's password. The app gets an access token with limited scopes; the user approves the access.

WHY IS IT IMPORTANT?

> Integrations with Google, Microsoft, Slack and CRMs use OAuth. Mistakes leak data.

WHERE IS IT USED IN INDUSTRY?

Every SaaS integration.

HOW DOES IT WORK?

- Authorisation code flow with PKCE for apps.
- Scopes limit what the token can do.
- Tokens expire; refresh tokens renew them.
- Never put tokens in URLs or logs.

PRACTICAL EXAMPLE

> A Slack bot asks for chat:write scope only; the token is stored in a secrets manager.

TOOL OR SOFTWARE

Postman OAuth helper

HANDS-ON EXERCISE

Get an OAuth token from a sandbox provider and call one endpoint with it.

COMMON MISTAKES

Using long-lived tokens in code.

INTERVIEW QUESTION

What is the difference between authentication and authorisation in OAuth?

OFFICIAL AND FURTHER READING

- OAuth 2.0 simplified (learning site)

B8.3

MCP security and business integration examples

WHAT IS IT?

> MCP servers expose tools to AI clients. Security needs scoped permissions, user consent for sensitive tools, and logs. Integration examples cover CRM, ERP, Jira, Slack and Notion.

WHY IS IT IMPORTANT?

> A badly scoped MCP tool can change business data.

WHERE IS IT USED IN INDUSTRY?

AI-enabled business tools in every sector.

HOW DOES IT WORK?

- Expose read tools first; add write tools with approval.
- Scope each tool to one system and user role.
- Log every tool call.

PRACTICAL EXAMPLE

> An MCP server lets the assistant read Jira tickets but needs approval to close them.

TOOL OR SOFTWARE

MCP documentation (official)

HANDS-ON EXERCISE

Design an MCP tool list for a support team with read and write tools marked.

COMMON MISTAKES

Giving an AI client admin tokens.

INTERVIEW QUESTION

How would you secure an MCP server that writes to a CRM?

OFFICIAL AND FURTHER READING

- Model Context Protocol (official)

B9 Linux and the Command Line (Part C gap)

B9.1

Linux and terminal essentials for data work

WHAT IS IT?

> Linux is the server operating system most data platforms run on. The command line lets you manage files, processes and services.

WHY IS IT IMPORTANT?

> Cloud servers, containers and CI runners are Linux; you need the terminal to debug them.

WHERE IS IT USED IN INDUSTRY?

Data engineering and DevOps roles.

HOW DOES IT WORK?

- Navigate: pwd, ls, cd, find, grep.
- Files: cat, head, tail -f, cp, mv, chmod.
- Processes: ps, top, kill.
- Pipes and redirection: |, >, >>.

PRACTICAL EXAMPLE

> tail -f pipeline.log | grep ERROR shows errors live as the pipeline runs.

TOOL OR SOFTWARE

Any Linux or WSL terminal

HANDS-ON EXERCISE

Find all ERROR lines in a log file from the last 1,000 lines and count them.

COMMON MISTAKES

Running commands from unknown sources.

INTERVIEW QUESTION

How would you find which process is using port 8080?

OFFICIAL AND FURTHER READING

- The Missing Semester (MIT, free)
- Linux command reference (free book)

B10 Technical Documentation and Production Troubleshooting (Part C gap)

B10.1

Technical documentation and runbooks

WHAT IS IT?

> Documentation explains what a system does, how to run it and how to fix it. A runbook is a step-by-step guide for common incidents.

WHY IS IT IMPORTANT?

> Handover and on-call work fail without documentation.

WHERE IS IT USED IN INDUSTRY?

Required in every production team.

HOW DOES IT WORK?

- README: purpose, setup, run, owners.
- Data dictionary: tables, columns, meanings.
- Runbook: symptom, check, fix, escalation.

PRACTICAL EXAMPLE

> A runbook for 'dashboard shows yesterday's numbers': check the pipeline run, re-run the failed task, notify the owner.

TOOL OR SOFTWARE

Markdown, Confluence

HANDS-ON EXERCISE

Write a runbook for one failure in your project.

COMMON MISTAKES

Documentation written once and never updated.

INTERVIEW QUESTION

What should a runbook contain?

OFFICIAL AND FURTHER READING

- Google SRE book: effective troubleshooting (free)

B10.2

Production troubleshooting for data pipelines and AI apps

WHAT IS IT?

> Troubleshooting finds why a live system fails: check what changed, read logs, reproduce, fix, verify and review.

WHY IS IT IMPORTANT?

> Stakeholders judge data teams on how fast and calmly they recover.

WHERE IS IT USED IN INDUSTRY?

On-call data engineers and FDEs.

HOW DOES IT WORK?

- Confirm impact and mitigate first (rerun, roll back, switch source).
- Check recent changes, logs and upstream freshness.
- Fix the root cause and add a test or alert.
- Write a short post-incident review.

PRACTICAL EXAMPLE

> A dashboard is wrong: an upstream schema change renamed a column; the fix maps the new name and a schema-change alert is added.

TOOL OR SOFTWARE

Airflow logs, dbt docs, cloud logs

HANDS-ON EXERCISE

Inject a broken schema change in a sample pipeline and write the incident note.

COMMON MISTAKES

Fixing without finding the cause.

INTERVIEW QUESTION

A pipeline failed at 3 am. What do you do first?

OFFICIAL AND FURTHER READING

- PagerDuty incident response guide (official)